SAGN: Sharpening-Aware Graph Network for Hyperspectral Image Change Detection

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***Abstract*— Graph neural networks (GNNs) have garnered significant attention in hyperspectral image (HSI) change detection (CD). However, existing GNN-based methods extract features by aggregating neighborhood information, which is essentially a low-pass Laplacian smoothing operation and tends to diminish change information between bitemporal HSIs. In addition, these methods rely on fixed hand-crafted graphs, and thus cannot capture complex structures of HSIs well. To address these deficiencies, this article develops a sharpening-aware graph network (SAGN) for achieving high-quality HSI CD. First, to counteract the weakening of differences caused by Laplacian smoothing, this article proposes a novel Laplacian sharpening-based graph convolution (LSGC) module to accentuate change information between bitemporal HSIs. Second, instead of using "similarity graphs," this article constructs untied "difference graphs" for bitemporal HSIs to model dissimilarities between changed pixels and their neighbors. The SAGN can dynamically update graph structures across all layers, aiming to further maximize the divergence. Finally, a joint loss function, incorporating modified cross-entropy loss and contrastive loss, is devised to enhance interclass discrimination of learned features and alleviate the issues stemming from imbalanced labeled samples. Experiments on various HSI CD datasets demonstrate the effectiveness and superiority of the proposed SAGN.**

***Index Terms*— Graph convolutional network (GCN), hyperspectral image (HSI) change detection (CD), Laplacian sharpening.**

I. Introduction

HYPERSPECTRAL image (HSI) change detection (CD) stands as a prominent area of interest within remote sensing [1], [2], [3]. Given bitemporal HSIs (i.e., two HSIs collected from the same area under two different time periods), the primary objective of HSI CD is to pinpoint and identify the changed regions between these bitemporal HSIs. HSIs encompass a wealth of spectral and spatial distribution information related to land cover, facilitating the precise identification of subtle changes in Earth observation scenarios [4], [5], [6], [7]. Consequently, HSI CD has emerged as a competitive technology with diverse applications across domains like urban planning, agricultural analysis, and environmental monitoring [8], [9], [10], [11].

Early conventional methods for HSI CD primarily relied on measuring spectral differences between bitemporal HSIs to gauge the extent of change [12]. Among these, prominent approaches included change vector analysis (CVA), multivariate change detection (MAD), support vector machine (SVM), and others [13], [14], [15]. CVA typically subtracts HSIs to derive spectral change vectors, with their magnitudes serving as indicators of change degrees. Subsequently, the norms of change vectors are classified using a threshold-based method. MAD employs canonical correlation analysis to maximize the correlation between bitemporal HSIs, and then subtracts HSIs followed by a threshold-based process to generate the change map. SVM combines bitemporal HSIs by concatenating them along the spectral dimension. Then, the concatenated spectral signatures are classified into changed or unchanged categories. While these methods aim to enhance and further mine change information between HSIs, and always have clear practical meanings. However, these methods rely solely on low-level spectral differences and cannot fully exploit the inherent characteristics of HSIs.

In recent years, deep learning (DL) has exhibited remarkable potential in HSI CD task [16]. Leveraging its powerful feature learning capabilities, DL-based methods offer distinct advantages in processing bitemporal HSIs and have demonstrated significant superiority in HSI CD. Initially, many DL-based methods are tailored to concentrate primarily on spectral signatures of HSIs and leverage rich spectral information [17]. Subsequently, massive works apply convolution neural networks (CNNs) to exploit spatial-spectral information of bitemporal HSIs [18], [19], [20]. For instance, Ou et al. [18] introduced a CNN framework employing slow-fast band selection and feature fusion grouping methods to effectively differentiate changed and unchanged pixels. In [19], a compact band weighting strategy and multiscale attention-based method was developed to leverage multilevel spatial information and prioritize pixels to be classified. Luo et al. [20] proposed a multiscale diff-changed feature fusion network that captures change components between bitemporal HSIs by extracting

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and fusing changed features across various scales, highlighting subtle changes. In [21], a deep multiscale pyramid network was proposed to extract features at multiple levels and scales for HSI CD. Yang et al. [22] proposed an untrained model using a randomly initialized method with a spatial-channel augmentation module for HSI CD. To enhance the exploration of temporal correlations between HSIs, several methods incorporated transformer [23] into HSI CD frameworks to improve detection performance. For instance, Wang et al. [24] proposed a joint spectral, spatial, and temporal transformer to sequentially extract spectral sequence information, spatial texture information, and temporal details from HSIs. Zhang et al. [25] introduced a cascade spectral-aware transformer designed to exploit sequence properties in spectral features, incorporating a spectral constraint module to reduce spectral distortion. Additionally, to address the limitations of patchwise mode, a triplet transformer framework [26] was proposed. This framework extracts temporal change information and utilizes middle- and long-range dependencies for improved perception. Although these HSI CD methods have made considerable advancements, they still suffer from some drawbacks. To be specific, these methods operate solely in the patchwise mode, limiting their ability to capture comprehensive global interactive information across the entire HSI. Moreover, in order to align with the patchwise mode, these methods require dividing HSIs into overlapping HSI cubes in advance and then conduct the convolution on these overlapping HSI cubes individually. However, this approach conducts a lot of replication computation, resulting in prolonged training time.

Recent advancements in graph neural networks (GNNs) have garnered significant attention and demonstrated remarkable potential in the field of HSI CD [27], [28], [29], [30]. To achieve high-quality detection performance, a variety of GNN frameworks, e.g., graph convolutional networks (GCNs) [31] and graph attention networks [32], have been leveraged in HSI CD and proved to be effective. For instance, Qu et al. [33] introduced a dual-branch difference amplification GCN (D^2AGCN) aimed at extracting graph representations with amplified difference information, and achieved satisfactory performance in HSI CD. Wang et al. [34] proposed a dual-branch framework based on temporal-spatial joint graph attention with complement strategy (CSDBF) to aggregate temporal and spatial information from bitemporal HSIs. Dong et al. [35] introduced a dual–branch local information-enhanced graph-transformer (D-LIEG) designed to extract local-global spectral–spatial features from bitemporal HSIs. Zhang et al. [36] designed a multiorder GCN for HSI CD to utilize the multiorder information among graph nodes. Qu et al. [37] proposed a GCN-based domain adaptive framework to improve the feature expression capability and generalization.

Despite the advancements, several significant issues persist in these GNN-based methods. First, existing methods generally extract features by aggregating neighborhood information, which is essentially a Laplacian smoothing-based graph convolution. However, the low-pass smoothing operation makes the features of changed and unchanged pixels similar, and reduces differences in changed regions. Second, existing methods often construct a single "similarity graph" by computing similarities between concatenated spectral signatures, and use it to guide information aggregation in both bitemporal HSIs, which overlooks the distinct nature of bitemporal HSIs. More importantly, graph structures are always noisy due to complex spectral variability, but cannot be adaptively refined during the training, thereby limiting the efficacy. Moreover, practical HSI CD datasets exhibit a significant imbalance between changed and unchanged pixels. This poses a challenge to GNN-based methods, as the training process is easily dominated by unchanged pixels.

To tackle the aforementioned challenges, this article proposes a novel framework termed the sharpening-aware graph network (SAGN) for HSI CD. First, distinct from the low-pass graph convolution-based methods, we propose a novel Laplacian sharpening-based graph convolution (LSGC) module to accentuate change information between bitemporal HSIs by pushing the features of changed samples farther away from those of their neighbors. Second, we construct untied "difference graphs" to model variations between changed samples and their neighbors, and also dynamically update graph structures during training to make them more compatible with the LSGC module. It is important to highlight that we construct different "difference graphs" tailored to bitemporal HSIs, instead of using a tied graph to guide the graph convolutions. Finally, a joint loss function is developed by integrating the modified cross-entropy loss and the contrastive loss. The modified cross-entropy loss is intended to allocate larger weights to changed pixels, whereas the contrastive loss aims to distinctly differentiate the change probabilities between changed and unchanged pixels. The contrastive loss ensures a clear margin between the probabilities, facilitating their discernibility.

The SAGN distinguishes itself from existing methods in following crucial aspects. First, we prioritize the significance of difference information between changed pixels and their neighbors, and propose an LSGC to enhance the separability of change regions. This contrasts with existing methodologies that typically employ Laplacian smoothing-based graph convolution, which tends to homogenize the feature representations of change pixels with their neighboring pixels, leading to the loss of crucial difference information. Second, we construct "difference graphs" and dynamically refine graph structures to enable the learned graphs to model difference relations well. In contrast, other methods typically construct "similarity graphs" to guide information aggregation, and often employ fixed graph structures or only focus on learning graph structures based on feature similarity, neglecting the crucial distinction between difference and similarity information.

The primary contributions can be outlined as follows.

1) A novel SAGN framework is proposed for highly precise HSI CD. Through the LSGC module and the dynamic graph strategy, the SAGN can substantially boost the difference information within change regions. By developing a joint loss function, the SAGN effectively mitigates the challenge posed by class imbalance. The SAGN showcases superior performance on classic HSI CD datasets.

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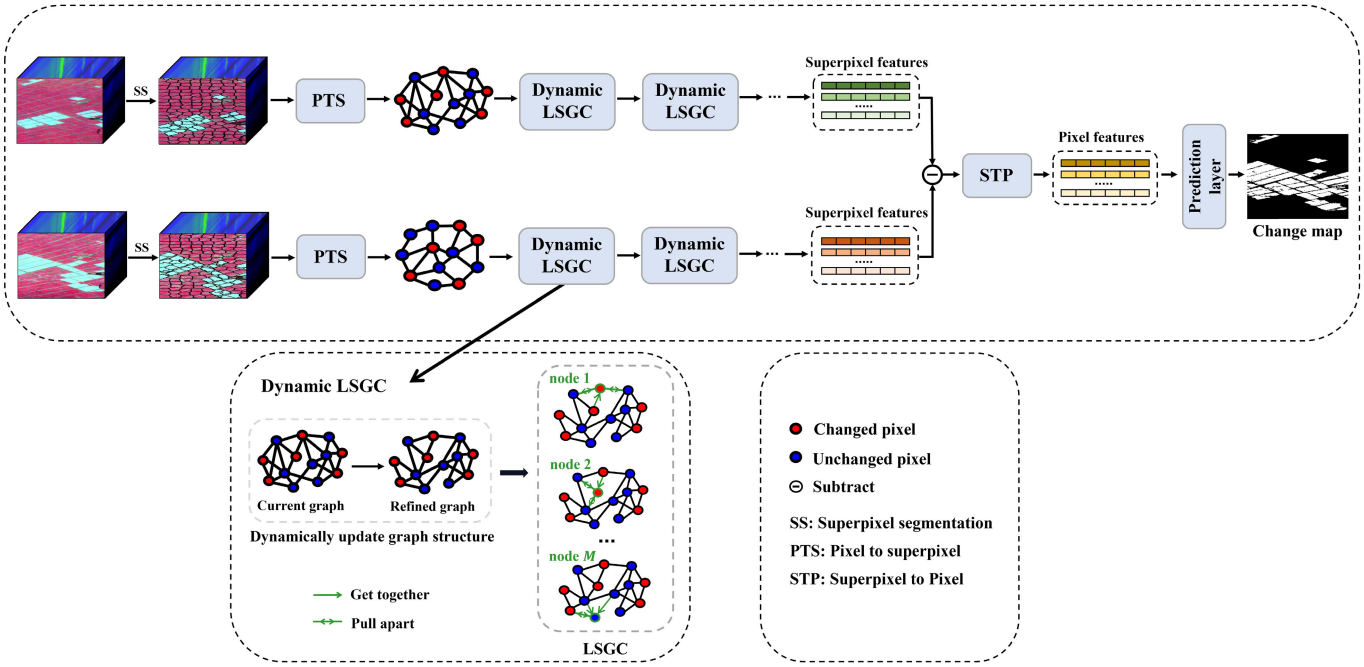


Fig. 1. Flowchart of the proposed SAGN. The SAGN first converts the bitemporal HSIs into two untied "difference graphs." Then, multiple dynamic LSGCs are utilized to dynamically update graph structures and extract discriminative features. The LSGC module is a Laplacian sharpening operation, which is able to push the features of changed samples far away from those of their neighbors. Finally, the SAGN projects superpixel features back into the HSI space, and generates CD results.

2) An LSGC module with Laplacian sharpening operation is built to significantly enhance and emphasize changed pixels. The Laplacian sharpening can force changed pixels away from their neighboring pixels, and thus accentuates change information between bitemporal HSIs.
3) A dynamic graph strategy is proposed to improve the adaptability of SAGN. Instead of "similarity graphs," we construct "difference graphs" to model neighborhood relations. More importantly, "difference graph" structures are dynamically updated across all layers to align with the SAGN.

The subsequent sections of this article are organized as follows. Section II provides a detailed exposition of the proposed SAGN. Following that, Section III showcases the experimental results on various HSI CD datasets. Finally, Section IV concludes this article.

II. Sharpening-Aware Graph Network

In this section, the details of the proposed SAGN are provided. As the pipeline of SAGN presented in Fig. 1, the SAGN first segments the bitemporal HSIs into a number of superpixels by superpixel segmentation (SS), and obtains superpixel features by the pixel to superpixel (PTS) operation. Then, the SAGN constructs two untied "difference graphs" for bitemporal HSIs, respectively. Next, multiple dynamic LSGCs are utilized to dynamically update graph structures and extract discriminative features. The LSGC module is a Laplacian sharpening operation, which is able to push the features of changed samples far away from those of their neighbors, thus significantly enhancing change information between bitemporal HSIs. Finally, the pixel features are derived by projecting the learned feature representations back into the HSI space, which are then utilized to generate CD results. The critical operations within the proposed SAGN will be expounded upon by presenting the LSGC module (see Section II-C), describing the dynamic LSGC (see Section II-D), elaborating the SAGN-based CD process (see Section II-E), introducing the proposed joint loss and giving the optimization algorithm (see Section II-F).

A. Notations

This section gives some notations utilized within this article. Let $\bar{\boldsymbol{Z}}_{\mathcal{T}_1} \in \mathbb{R}^{H\times W\times B}$ and $\bar{\boldsymbol{Z}}_{\mathcal{T}_2} \in \mathbb{R}^{H\times W\times B}$ represent bitemporal HSIs of the same size, where H, W, and B denote the height, width, and number of spectral bands, respectively. The set of class labels is denoted by $\mathcal{C} = \{0, 1\}$, where 1 represents changed pixel pair and 0 represents unchanged pixel pair. In this article, $\mathcal{D}_L = \{(\bar{\boldsymbol{z}}_{\mathcal{T}_1,l_i}, \bar{\boldsymbol{z}}_{\mathcal{T}_2,l_i}, y_i)\}_{i=1}^{N}$ represents the labeled set consisting of N labeled pixel pairs, where $\bar{\boldsymbol{z}}_{\mathcal{T}_1,l_i} \in \mathbb{R}^B$ and $\bar{\boldsymbol{z}}_{\mathcal{T}_2,l_i} \in \mathbb{R}^B$ are the spectral signatures of the ith labeled pixel pair, and $y_i \in \mathcal{C}$ represent its label.

B. Superpixel-Based Graph Construction

Taking account into the computational efficiency and the spectral–spatial correlation between pixels in HSIs, the proposed method operates on superpixel-based graphs rather than pixel-based graphs. To this end, we initially segment each of the bitemporal HSIs into many spatially connected superpixels based on the spectral-spatial similarity, and then transform

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these superpixels into a superpixel-based graph. The same SS method and graph construction method are employed for both bitemporal HSIs. Therefore, for illustration purposes, we will use one of the bitemporal HSIs as an example, namely $\bar{\boldsymbol{Z}}$ ($\bar{\boldsymbol{Z}} = \bar{\boldsymbol{Z}}_{\mathcal{T}_1}$ or $\bar{\boldsymbol{Z}} = \bar{\boldsymbol{Z}}_{\mathcal{T}_2}$), to show this process. The detailed procedures are outlined as follows.

1) Superpixel Segmentation: First, the linear discriminant analysis is applied to reduce the dimension of the bitemporal HSIs. Then, the simple linear iterative clustering (SLIC) method [38] is utilized to partition bitemporal HSIs into numerous spatially connected and spectrally similar superpixels. Specifically, we segment the HSI $\bar{\boldsymbol{Z}}$ into $M = \lceil (H \times W)/\mu \rceil$ superpixels, where $\mu (1 \leq \mu)$ is a scale hyperparameter in SLIC, governing the average size of superpixels.

a) Pixel to superpixel: After segmentation, the spectral feature of each superpixel is determined by averaging the spectral signatures of all pixels within this superpixel. To facilitate this process, we establish a mapping association between pixels and superpixels. Specifically, let $\hat{\boldsymbol{Z}} = \text{flatten}(\bar{\boldsymbol{Z}}) \in \mathbb{R}^{HW \times B}$ denote the flattened HSI, let $\mathcal{S} = \{\mathcal{S}_i\}_{i=1}^{M}$ denote the superpixel set and $\mathcal{S}_i$ denote the ith superpixel, where $\mathcal{S}_i \cap \mathcal{S}_j = \varnothing (\forall i \neq j)$ and $H \times W = \sum_{i=1}^{M} N_i$. And the association matrix $\boldsymbol{Q} \in \mathbb{R}^{HW \times M}$ can be defined as

$$\boldsymbol{Q}_{i,j} = \begin{cases} 1, & \text{if } \hat{\boldsymbol{z}}_i \in \mathcal{S}_j \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $\hat{\boldsymbol{z}}_i$ is the ith pixel in $\hat{\boldsymbol{Z}}$ and $\mathcal{S}_j$ is the jth superpixel. Then, $\boldsymbol{Q}$ is column-normalized to derive the normalized mapping association $\hat{\boldsymbol{Q}}$ by calculating $\hat{\boldsymbol{Q}}_{i,j} = \boldsymbol{Q}_{i,j} / \sum_m \boldsymbol{Q}_{m,j}$. After that, the spectral feature matrix of all superpixels can be simply obtained as

$$\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_M]^T = \hat{\boldsymbol{Q}}^T \hat{\boldsymbol{Z}} \triangleq \text{PTS}(\hat{\boldsymbol{Z}}) \tag{2}$$

where $\boldsymbol{X} \in \mathbb{R}^{M \times B}$. Utilizing the transformation described in (2), the HSI can be easily encoded into graph nodes, subsequently allowing processing with the GCN.

b) Superpixel to pixel (STP): Since our goal is to detect changed pixels rather than superpixels, the features of superpixels should be assigned to pixels before detection. Specifically, let $\hat{\boldsymbol{X}} \in \mathbb{R}^{M \times \hat{B}}$ denote the learned features of all superpixels, then the learned features of all pixels can be obtained as

$$\hat{\boldsymbol{Z}}^* = \left[\hat{\boldsymbol{z}}_1^*, \hat{\boldsymbol{z}}_2^*, \ldots, \hat{\boldsymbol{z}}_{HW}^*\right]^T = \boldsymbol{Q}\hat{\boldsymbol{X}} \triangleq \text{STP}(\hat{\boldsymbol{X}}) \tag{3}$$

where $\hat{\boldsymbol{Z}}^* \in \mathbb{R}^{HW \times \hat{B}}$. After this, the learned features of all pixels can be projected back to the HSI space by $\bar{\boldsymbol{Z}}^* = \text{reshape}(\hat{\boldsymbol{Z}}^*) \in \mathbb{R}^{H \times W \times \hat{B}}$, where reshape$(\cdot)$ is the operation of restoring the spatial dimension of the flattened pixel feature.

2) Graph Construction: Once the HSI has been partitioned into M superpixels, we can transform these superpixels into a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E}$ denote the node set and edge set. Specifically, in this article, $\mathcal{V}$ is composed of all superpixels, and $\mathcal{E}$ contains all edges that spatially connect two superpixels.

Each superpixel node $v_i \in \mathcal{V} (i = 1, 2, \ldots, M)$ is associated with a spectral feature $\boldsymbol{x}_i \in \mathbb{R}^B$. Thus, the initial feature matrix of the graph is $\boldsymbol{X}$ obtained above. Let $\boldsymbol{E} \in \mathbb{R}^{M \times M}$ denote the edge matrix of $\mathcal{G}$, where each element represents the presence of an edge between each pair of superpixel nodes, which is given by

$$(\boldsymbol{E})_{ij} = \begin{cases} 1, & \text{if } v_i \text{ and } v_j \text{ are spatially adjacent} \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where v_i and v_j are spatially adjacent if they share a common boundary.

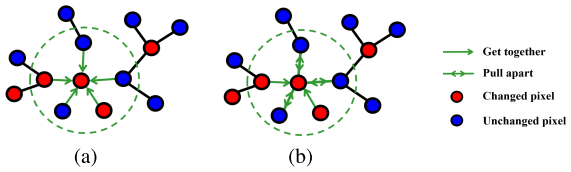


Fig. 2. Comparison of Laplacian sharpening and Laplacian smoothing. (a) Laplacian smoothing-based graph convolution. (b) LSGC.

C. Laplacian Sharpening-Based Graph Convolution

Li et al. [39] demystify GCN and show that GCN can be understood as a special form of Laplacian smoothing. It is known that Laplacian smoothing refers to a process that computes a new feature of the input by taking a weighted average of its neighbors and itself [40]. Therefore, Laplacian smoothing contradicts the goal of HSI CD, as the smoothing operation tends to homogenize the features of changed pixels, which inevitably weakens the change information between bitemporal HSIs.

In contrast to existing studies, this article adopts Laplacian sharpening, a counterpart of Laplacian smoothing, which is a process that pushes the feature representation of changed pixels further apart from their neighbors. To this end, we utilize a typical type of Laplacian sharpening [40]. Specifically, assume there are M nodes in a superpixel graph, $\boldsymbol{X}^{\text{in}} \in \mathbb{R}^{M \times d}$ denotes the corresponding feature matrix, $\tilde{\boldsymbol{A}} \in \mathbb{R}^{M \times M}$ is the adjacency matrix, and $\tilde{\boldsymbol{D}} = (\tilde{D}_{ii})$ represents the degree matrix of $\tilde{\boldsymbol{A}}$, where $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$. Then, the graph convolution based on the Laplacian sharpening is expressed as

$$\boldsymbol{X}^{\text{out}} = \varphi\left(\left(2\boldsymbol{I} - \tilde{\boldsymbol{D}}^{-1/2}\tilde{\boldsymbol{A}}\tilde{\boldsymbol{D}}^{-1/2}\right)\boldsymbol{X}^{\text{in}}\boldsymbol{W}\right) \tag{5}$$

where $\boldsymbol{W}$ is a trainable weight matrix, $\boldsymbol{I}$ denotes the identity matrix, $\varphi(\cdot)$ is the rectified linear unit (ReLU) activation function, and $2\boldsymbol{I} - \tilde{\boldsymbol{D}}^{-1/2}\tilde{\boldsymbol{A}}\tilde{\boldsymbol{D}}^{-1/2}$ is a special form of Laplacian sharpening.

Nevertheless, the spectral radius of $2\boldsymbol{I} - \tilde{\boldsymbol{D}}^{-1/2}\tilde{\boldsymbol{A}}\tilde{\boldsymbol{D}}^{-1/2}$ is 3, and iteratively applying this operator will be numerically instable. Hence, similar to GCN, we also want to find a numerically stable form of Laplacian sharpening in (5). Specifically, we introduce the following renormalization trick:

$$2\boldsymbol{I} - \tilde{\boldsymbol{D}}^{-1/2}\tilde{\boldsymbol{A}}\tilde{\boldsymbol{D}}^{-1/2} \rightarrow \hat{\boldsymbol{D}}^{-1/2}\hat{\boldsymbol{A}}\hat{\boldsymbol{D}}^{-1/2} \tag{6}$$

where $\hat{\boldsymbol{A}} = 2\boldsymbol{I} - \tilde{\boldsymbol{A}}$ and $\hat{\boldsymbol{D}} = 2\boldsymbol{I} + \tilde{\boldsymbol{D}}$. The intuitive comparison between Laplacian sharpening and Laplacian smoothing is shown in Fig. 2.

In summary, the numerically stable LSGC module is given as

$$\boldsymbol{X}^{\text{out}} = \varphi\left(\left(\hat{\boldsymbol{D}}^{-1/2}\hat{\boldsymbol{A}}\hat{\boldsymbol{D}}^{-1/2}\right)\boldsymbol{X}^{\text{in}}\boldsymbol{W}\right) \tag{7}$$

where $\hat{\boldsymbol{A}} = 2\boldsymbol{I} - \tilde{\boldsymbol{A}}$ and $\hat{\boldsymbol{D}} = 2\boldsymbol{I} + \tilde{\boldsymbol{D}}$.

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D. Dynamic LSGC

The efficacy of graph convolution heavily depends on the quality of the predefined graph. However, as previously mentioned, existing methods commonly use the graph to encode the similarities and generally perform Laplacian smoothing-based graph convolution to aggregate neighboring information, which will make the feature representations of changed pixels undistinguishable with those of their neighboring pixels.

To address this weakness, we first construct a learnable "difference graph" to capture the dissimilarities between pixels and their neighbors, and then push the features of change pixels far away from those of neighbors by utilizing this constructed "difference graph" as a guide. Specifically, by introducing a trainable weight matrix denoted as $\boldsymbol{W}_\phi$, the learnable difference graph constructed in this work can be formulated as

$$\boldsymbol{A} = \left(\mathbf{1} - \text{Sigmoid}\left((\boldsymbol{X}\boldsymbol{W}_\phi)(\boldsymbol{X}\boldsymbol{W}_\phi)^T\right)\right) \odot \boldsymbol{E} \tag{8}$$

where Sigmoid$(\cdot)$ represents the sigmoid function and $\odot$ denotes the Hadamard product. In (8), the input $\boldsymbol{X}$ is first transformed to $\bar{\boldsymbol{X}} = \boldsymbol{X}\boldsymbol{W}_\phi$. Then, the similarities between all node pairs are calculated by $\bar{\boldsymbol{S}} = \text{Sigmoid}(\bar{\boldsymbol{X}}\bar{\boldsymbol{X}}^T)$, where the sigmoid function compresses the similarities to a range of $(0, 1)$. Next, the differences between all node pairs are obtained by $\boldsymbol{S} = \mathbf{1} - \bar{\boldsymbol{S}}$. Finally, the edge mask $\boldsymbol{E}$ is applied to remove the associations between nonadjacent nodes, resulting in $\boldsymbol{A} = \boldsymbol{S} \odot \boldsymbol{E}$.

Since the features are dynamically updated along with the graph convolution, utilizing a single trainable matrix $\boldsymbol{W}_\phi$ is inadequate to adaptively capture dissimilarities among nodes across all layers. Thus, we intend to dynamically learn graph structures across all layers. Specifically, we use multiple trainable matrixes $\boldsymbol{W}_\phi^{(l)} \in \mathbb{R}^{n\times n}$ (n denotes the feature dimension, and $l = 1, 2, \ldots, L$, where L denotes the number of layers) to dynamically represent dissimilarities. Then, (8) can be refined as

$$\boldsymbol{A}^{(l)} = \left(\mathbf{1} - \text{Sigmoid}\left(\left(\boldsymbol{H}^{(l-1)}\boldsymbol{W}_\phi^{(l)}\right)\left(\boldsymbol{H}^{(l-1)}\boldsymbol{W}_\phi^{(l)}\right)^T\right)\right) \odot \boldsymbol{E} \tag{9}$$

where $\boldsymbol{W}_\phi^{(l)}$ is a trainable weight matrix and $\boldsymbol{A}^{(l)}$ is the dynamic difference graph in the lth layer.

With the dynamic difference graph, we can build the dynamic LSGC. Assume there are L layers in the dynamic LSGC and n units in the lth layer ($l = 1, 2, \ldots, L$), and then the lth layer of dynamic LSGC can be formulated as

$$\boldsymbol{H}^{(l)} = \varphi\left(\left(\left(\hat{\boldsymbol{D}}^{(l)}\right)^{-1/2}\hat{\boldsymbol{A}}^{(l)}\left(\hat{\boldsymbol{D}}^{(l)}\right)^{-1/2}\right)\boldsymbol{H}^{(l-1)}\boldsymbol{W}^{(l)}\right) \tag{10}$$

where $\hat{\boldsymbol{A}}^{(l)} = 2\boldsymbol{I} - \boldsymbol{A}^{(l)}$, $\hat{\boldsymbol{D}}^{(l)} = 2\boldsymbol{I} + \boldsymbol{D}^{(l)}$, and $\boldsymbol{W}^{(l)} \in \mathbb{R}^{n\times n}$ is the learnable weight matrix, and $\boldsymbol{D}^{(l)}$ represents the degree matrix of $\boldsymbol{A}^{(l)}$ with $D_{ii}^{(l)} = \sum_j A_{ij}^{(l)}$. Before being fed into the dynamic LSGC, the input $\boldsymbol{X} \in \mathbb{R}^{M\times B}$ is first transformed to $\boldsymbol{H}^{(0)} \in \mathbb{R}^{M\times n}$ by $\boldsymbol{H}^{(0)} = \boldsymbol{X}\boldsymbol{W}^{(0)}$, where $\boldsymbol{W}^{(0)} \in \mathbb{R}^{B\times n}$ is a learnable weight matrix.

Algorithm 1 Dynamic LSGC

Input: Input feature $\boldsymbol{X}$.
Output: $\boldsymbol{H}^{(L)}$.

$\boldsymbol{H}^{(0)} = \boldsymbol{X}\boldsymbol{W}^{(0)}$;
for $l = 1, 2, \cdots, L$ **do**
  $\boldsymbol{A}^{(l)} = (\mathbf{1} - \text{Sigmoid}((\boldsymbol{H}^{(l-1)}\boldsymbol{W}_\phi^{(l)})(\boldsymbol{H}^{(l-1)}\boldsymbol{W}_\phi^{(l)})^T)) \odot \boldsymbol{E}$;
  $\hat{\boldsymbol{A}}^{(l)} = 2\boldsymbol{I} - \boldsymbol{A}^{(l)}$, $\hat{\boldsymbol{D}}^{(l)} = 2\boldsymbol{I} + \boldsymbol{D}^{(l)}$;
  $\boldsymbol{H}^{(l)} = \varphi\left(\left((\hat{\boldsymbol{D}}^{(l)})^{-\frac{1}{2}}\hat{\boldsymbol{A}}^{(l)}(\hat{\boldsymbol{D}}^{(l)})^{-\frac{1}{2}}\right)\boldsymbol{H}^{(l-1)}\boldsymbol{W}^{(l)}\right)$;
end

The proposed dynamic LSGC module is briefly recorded as

$$\boldsymbol{H}^{(L)} \triangleq \text{dynamicLSGC}(\boldsymbol{X}, L) \tag{11}$$

where $\boldsymbol{X}$ and $\boldsymbol{H}^{(L)}$ are the input and the output of the dynamic LSGC module, respectively. The procedures of the dynamic LSGC module are summarized in Algorithm 1.

E. SAGN-Based Change Detection

Given the bitemporal HSIs $\hat{\boldsymbol{Z}}_{\mathcal{T}_1} \in \mathbb{R}^{HW\times B}$ and $\hat{\boldsymbol{Z}}_{\mathcal{T}_2} \in \mathbb{R}^{HW\times B}$, we first use the superpixel-based graph construction (SGC) method described in Section II-B to transform them into the graph space and obtain two graphs $\mathcal{G}_{\mathcal{T}_1} = (\mathcal{V}_{\mathcal{T}_1}, \mathcal{E}_{\mathcal{T}_1})$ and $\mathcal{G}_{\mathcal{T}_2} = (\mathcal{V}_{\mathcal{T}_2}, \mathcal{E}_{\mathcal{T}_2})$, and the corresponding feature matrixes are obtained by

$$\begin{cases} \boldsymbol{X}_{\mathcal{T}_1} = \text{PTS}\left(\hat{\boldsymbol{Z}}_{\mathcal{T}_1}\right) \\ \boldsymbol{X}_{\mathcal{T}_2} = \text{PTS}\left(\hat{\boldsymbol{Z}}_{\mathcal{T}_2}\right) \end{cases} \tag{12}$$

where $\boldsymbol{X}_{\mathcal{T}_1} \in \mathbb{R}^{M_1\times B}$ and $\boldsymbol{X}_{\mathcal{T}_2} \in \mathbb{R}^{M_2\times B}$, here M_1 and M_2 represent the number of superpixels for the bitemporal HSIs $\hat{\boldsymbol{Z}}_{\mathcal{T}_1}$ and $\hat{\boldsymbol{Z}}_{\mathcal{T}_2}$, respectively.

The initial superpixel feature matrixes of $\boldsymbol{X}_{\mathcal{T}_1}$ and $\boldsymbol{X}_{\mathcal{T}_2}$ are feed into two different dynamic LSGC branches, which output two feature matrixes, i.e., $\boldsymbol{H}_{\mathcal{T}_1}^{(L)}$ and $\boldsymbol{H}_{\mathcal{T}_2}^{(L)}$. The formula is as follows:

$$\begin{cases} \boldsymbol{H}_{\mathcal{T}_1}^{(L)} = \text{dynamicLSGC}_{\mathcal{T}_1}(\boldsymbol{X}_{\mathcal{T}_1}, L) \\ \boldsymbol{H}_{\mathcal{T}_2}^{(L)} = \text{dynamicLSGC}_{\mathcal{T}_2}(\boldsymbol{X}_{\mathcal{T}_2}, L). \end{cases} \tag{13}$$

Next, we obtain the pixel features of bitemporal HSIs by (3), namely,

$$\begin{cases} \boldsymbol{Z}_{\mathcal{T}_1} = \text{STP}\left(\boldsymbol{H}_{\mathcal{T}_1}^{(L)}\right) \\ \boldsymbol{Z}_{\mathcal{T}_2} = \text{STP}\left(\boldsymbol{H}_{\mathcal{T}_2}^{(L)}\right). \end{cases} \tag{14}$$

The resulting $\boldsymbol{Z}_{\mathcal{T}_1}$ and $\boldsymbol{Z}_{\mathcal{T}_2}$ are then subtracted, followed by a prediction layer to compute the change probability. More specifically, the subtracted features are obtained as

$$\boldsymbol{D} = \boldsymbol{Z}_{\mathcal{T}_1} - \boldsymbol{Z}_{\mathcal{T}_2}. \tag{15}$$

Finally, a prediction layer is used to generate the change possibility

$$\hat{\boldsymbol{Y}} = \text{Sigmoid}(\boldsymbol{D}\boldsymbol{W}_p) \tag{16}$$

where $\boldsymbol{W}_p \in \mathbb{R}^{n\times 1}$ is the weight matrix in the prediction layer.

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F. Joint Supervision of Modified Cross-Entropy Loss and Contrastive Loss

For input labeled pair $(\bar{z}_{\mathcal{T}_1,l_i}, \bar{z}_{\mathcal{T}_2,l_i})$, the SAGN generates a change possibility score $\hat{y}_i$. The goal is to match the ground truth label, i.e., $y_i = 1$ for changed pixel pair and $y_i = 0$ for unchanged pixel pair. Thus, the binary cross-entropy loss is used as the basic objective function

$$\mathcal{J}_{\text{cel}}(\boldsymbol{W}) = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(\hat{y}_i\right) + (1-y_i)\log\left(1-\hat{y}_i\right)\right] \quad (17)$$

where $\boldsymbol{W} = \{\boldsymbol{W}^{(0)}, \{\boldsymbol{W}^{(l)}, and \boldsymbol{W}_{\phi}^{(l)}\}_{l=1}^{L}\}$ denotes all the parameters of the SAGN, which encompasses both the network parameters of the SAGN and the trainable parameters that are used to learn the dynamic difference graphs.

However, due to the larger number of unchanged pixels than that of change pixels, the unchanged pixels always comprise the majority of the loss and tend to dominate the loss function during training. As a result, the model could easily neglect the information of changed pixels, and thus misclassifies these changed pixels.

To this end, inspired by the focal loss [41], we propose to modify the binary cross-entropy loss to focus training on changed pixels, especially misclassified ones. More formally, a tunable focusing factor is introduced to modify the conventional cross-entropy loss, and the special form is defined as

$$\mathcal{J}_{\text{mcel}}(\boldsymbol{W}) = -\frac{1}{N}\sum_{i=1}^{N}\left(1-\hat{y}_i^{y_i}\left(1-\hat{y}_i\right)^{(1-y_i)}\right)^{\gamma} \times \left[y_i \log\left(\hat{y}_i\right) + (1-y_i)\log\left(1-\hat{y}_i\right)\right] \quad (18)$$

where γ is empirically set to 2 in the experiments.

To further enhance the discriminative ability and generalization ability of the SAGN, it is expected that the predicted change possibility score of changed pixel is obviously bigger than that of unchanged pixel and there is a large margin between them. To this end, we introduce the contrastive loss [42] into the HSI CD. Specifically, if a pixel is a changed pixel, its change possibility score should be bigger than an up-margin τ; if a pixel is an unchanged pixel, its change possibility score should be smaller than a down-margin $\tau - m$, that is, there is at least a margin of m between scores of changed and unchanged pixels. The formulation can be represented as

$$\mathcal{J}_{\text{cl}}(\boldsymbol{W}) = \frac{1}{N}\sum_{i=1}^{N}\left\{y_i\left[\tau-\hat{y}_i\right)]_+ + (1-y_i)\left[\hat{y}_i-(\tau-m)\right]_+\right\} \quad (19)$$

where $[\cdot]_+ = \max(0, \cdot)$.

Finally, a joint loss is proposed to supervise the final change possibility, defined as

$$\mathcal{J}_{\text{joint}}(\boldsymbol{W}) = \mathcal{J}_{\text{mcel}}(\boldsymbol{W}) + \lambda\mathcal{J}_{\text{cl}}(\boldsymbol{W}) \quad (20)$$

where $\lambda > 0$ is to balance the modified cross-entropy loss and the contrastive loss. By using this joint loss, the SAGN can pay more attention on the changed pixels during the training process. To sum up, the specifics of the training algorithm for the SAGN are summarized in Algorithm 2.

Algorithm 2 Training Algorithm of the SAGN

Input: $\hat{\boldsymbol{Z}}_{\mathcal{T}_1}$, $\hat{\boldsymbol{Z}}_{\mathcal{T}_2}$, hyperparameters λ, τ, and m, learning rate η, maximal iterations T_{max}.
Output: $\boldsymbol{W}$.

Set $t = 1$, randomly initialize all weights;
Transform $\hat{\boldsymbol{Z}}_{\mathcal{T}_1}$ and $\hat{\boldsymbol{Z}}_{\mathcal{T}_2}$ into the graph space, and obtain $\boldsymbol{X}_{\mathcal{T}_1}$ and $\boldsymbol{X}_{\mathcal{T}_2}$ by Eq.(12);
repeat
  $\boldsymbol{H}_{\mathcal{T}_1}^{(L)} = \text{dynamicLSGC}_{\mathcal{T}_1}(\boldsymbol{X}_{\mathcal{T}_1}, L)$;
  $\boldsymbol{H}_{\mathcal{T}_2}^{(L)} = \text{dynamicLSGC}_{\mathcal{T}_2}(\boldsymbol{X}_{\mathcal{T}_2}, L)$;
  Transform $\boldsymbol{H}_{\mathcal{T}_1}^{(L)}$ and $\boldsymbol{H}_{\mathcal{T}_2}^{(L)}$ back into the HSI space, and obtain $\boldsymbol{Z}_{\mathcal{T}_1}$ and $\boldsymbol{Z}_{\mathcal{T}_2}$ by Eq.(14);
  $\boldsymbol{D} = \boldsymbol{Z}_{\mathcal{T}_1} - \boldsymbol{Z}_{\mathcal{T}_2}$;
  $\hat{\boldsymbol{Y}} = \text{Sigmoid}(\boldsymbol{D}\boldsymbol{W}_p)$;
  Calculate $\mathcal{J}_{\text{mcel}}(\boldsymbol{W})$ by Eq.(18);
  Compute $\mathcal{J}_{\text{cl}}(\boldsymbol{W})$ by Eq.(19);
  $\mathcal{J}_{\text{joint}}(\boldsymbol{W}) \leftarrow \mathcal{J}_{\text{mcel}}(\boldsymbol{W}) + \lambda\mathcal{J}_{\text{cl}}(\boldsymbol{W})$;
  $\boldsymbol{W} \leftarrow \boldsymbol{W} - \eta\frac{\partial\mathcal{J}_{\text{joint}}(\boldsymbol{W})}{\partial\boldsymbol{W}}$;
  $t \leftarrow t + 1$;
until $t > T_{max}$;

III. Experiment

A. Experiment Settings

1) Datasets: Experiments are evaluated on representative HSI CD datasets, i.e., Farmland, Santa Barbara, and Bay Area. These datasets are briefly introduced as follows.

The Farmland dataset was obtained over a farmland near the city of Yancheng, Jiangsu province, China. The bitemporal HSIs were acquired by EO-1 Hyperion sensor on May 3, 2006, and April 23, 2007, respectively. Both of the bitemporal HSIs comprises 155 spectral bands and 450 × 140 pixels. The main change areas in the reference ground truth are farmland.

The Santa Barbara dataset was recorded by the AVIRIS sensor in 2013 and 2014 over Santa Barbara, CA, USA. The two HSIs have the same size of 984 × 740 × 224. This dataset contains 52 134 changed pixels, 80 418 unchanged pixels, and 595 608 unknown pixels.

The Bay Area dataset was collected in 2013 and 2015 by the AVIRIS sensor surrounding the city of Patterson in CA. The size of both bitemporal HSIs is 600 × 500 × 224. This dataset has 38 425 changed pixels, 34 211 unchanged pixels, and 227 364 unknown pixels.

2) Comparison Methods: To assess the performance, the proposed SAGN is compared against classical HSI CD methods, namely the CVA, the MAD, the SVM, and also state-of-the-art GCN-based methods, namely the GCN, the D^{2}AGCN [33], the CSDBF [34], and the D-LIEG [35]. All the compared methods utilize their default hyperparameter configurations. Besides, the SAGN is also compared with its variant named SAGN$_{\text{base}}$. Specifically, SAGN$_{\text{base}}$ only utilizes the LSGC-based network framework and is training supervised by the conventional cross-entropy loss.

3) Parameter Setting: On all datasets, the parameter settings are consistent, i.e., $\lambda = 0.2$, $L = 2$, $n = 128$ $\tau = 0.9$, $m = 0.8$. Furthermore, the learning rate η is set to 0.0005,

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TABLE I
CD PERFORMANCE ON THE SANTA BARBARA DATASET WITH 1% TRAINING SAMPLES

Method	OA(%)	AA(%)	$\kappa \times 100$	F1(%)	P(%)	R(%)
CVA	86.25±0.00	86.88±0.00	71.91±0.01	83.71±0.01	78.36±0.01	89.84±0.01
MAD	86.74±0.01	86.66±0.01	72.53±0.01	83.65±0.01	81.44±0.01	86.26±0.01
SVM	95.22±0.07	94.43±0.11	89.87±0.16	93.72±0.11	96.94±0.18	90.71±0.29
GCN	98.14±0.30	97.99±0.20	96.10±0.62	97.63±0.36	97.96±0.98	97.30±0.25
D^2AGCN	98.51±0.18	98.27±0.20	96.86±0.38	98.08±0.23	99.01±0.42	97.17±0.44
CSDBF	98.75±0.07	98.51±0.09	97.38±0.16	98.40±0.10	99.42±0.07	97.40±0.16
D-LIEG	98.41±0.07	98.24±0.13	96.66±0.15	97.97±0.10	98.53±0.34	97.42±0.46
$SAGN_{base}$	98.62±0.40	98.40±0.45	97.09±0.85	98.23±0.52	99.10±0.39	97.37±0.65
SAGN	**99.30±0.12**	**99.21±0.14**	**98.54±0.24**	**99.11±0.15**	**99.46±0.05**	**98.77±0.25**

while the number of training iterations T_{max} is set to 300. In the experiments, only 1% labeled samples are selected for training, while the remaining are used for testing. All experiments are performed on a computer with Intel Xeon W-2245 CPU and NVIDIA RTX A4000.

4) Evaluation Indicators: Overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) are the main indexes used for the quantitative evaluation, and precision (P), recall (R), and F1-score (F1) are also introduced as the auxiliary performance indexes [43]. Besides, true positive (TP) counts the number of changed pixels correctly classified into changed class, true negative (TN) counts the number of unchanged pixels correctly classified into unchanged class, false positive (FP) counts the number of unchanged pixels misclassified into changed class, and false negative (FN) counts the number of changed pixels misclassified into unchanged class. For these evaluation indexes, a larger value signifies superior CD performance. To mitigate bias, all evaluation indexes are computed by averaging the results from ten independent runs.

To visually depict the CD performance, a change map is also illustrated, where TP, TN, FP, and FN are, respectively, displayed as white pixels, black pixels, blue pixels, and red pixels.

B. Change Detection Performance

1) Santa Barbara Dataset: Table I displays the CD accuracies of all the comparison methods on the Santa Barbara dataset. The unsupervised learning methods, i.e., the CVA and the MAD, exhibit relatively lower performance in this challenging dataset, since it is difficult to distinguish the changed pixels and the unchanged pixels according to only spectral changes without leveraging labeled information. Conversely, as a conventional supervised learning method, the SVM achieves improved CD accuracy. This enhanced performance can be attributed to the supervised learning method's ability to learn from labeled data, enabling more effective discrimination between changed and unchanged pixels. Moreover, the spatial information plays a crucial role in HSI CD. Thus, GNN-based methods tend to achieve satisfactory accuracies as they effectively utilize spatial information among HSIs. This can be verified by the notable performance improvement observed in the GNN-based methods. After introducing the LSGC, the proposed SAGN have superior performance compared to other GNN-based methods that utilize unsuitable low-pass graph convolution. This indicates that the sharpening information is conducive for detecting changed regions of land covers. In contrast to $SAGN_{base}$, the accuracy of SAGN is notably improved, since the SAGN can dynamically consider difference relations among changed regions, thereby enabling better learning of subtle changing features. In general, our SAGN consistently surpasses other methods across all the performance indexes, demonstrating superior performance even under 1% of training ratio condition.

Fig. 3 presents the visual CD maps of all comparison methods on the Santa Barbara dataset. The conventional CVA and the MAD exhibit numerous FP pixels (blue) and FN pixels (red), which means they cannot distinguish changed pixels and unchanged pixels well. SVM also has a considerable number of misclassified pixels, since it only utilizes the low-level spectral features with limited discriminative capability for detecting subtle changes. The GNN-based methods, such as GCN, D^2AGCN, CSDBF, and D-LIEG, exhibit a higher count of FN pixels, which means they misclassify many changed pixels into unchanged class. In contrast, compared to other GNN-based methods, the proposed SAGN displays a distinct advantage in distinguishing detailed changes and correctly classifies most changed pixels, illustrated by the presence of fewer red pixels in the visual change maps. In general, our proposed SAGN exhibits the fewest FP pixels (blue) and FN pixels (red) among the compared methods, and thus achieves the best visual performance in the generated change maps.

2) Bay Area Dataset: Table II presents the CD accuracies on the Bay Area dataset. Compared to the CVA and MAD, other methods demonstrate superior OA and show improvements in terms of P and κ. This indicates that the CVA and MAD misclassify a large number of unchanged pixels into changed class, which is also evident from the visual maps (indicated by blue pixels), and thus have high R but low P.

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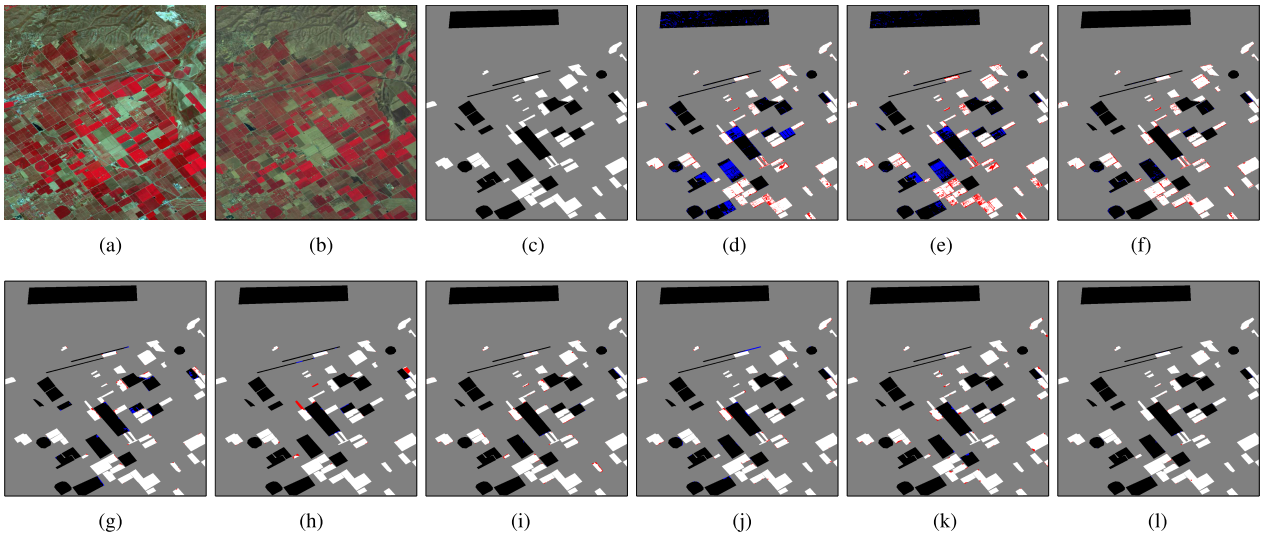


Fig. 3. CD maps of the compared methods on the Santa Barbara dataset. (a) False color image of Santa Barbara obtained in 2013. (b) False color image of Santa Barbara obtained in 2014. (c) Ground truth. (d) CVA, OA = 86.25%. (e) MAD, OA = 86.74%. (f) SVM, OA = 95.22%. (g) GCN, OA = 98.14%. (h) D^2AGCN, OA = 98.51%. (i) CSDBF, OA = 98.75%. (j) D-LIEG, OA = 98.41%. (k) $SAGN_{base}$, OA = 98.62%. (l) SAGN, OA = 99.30%. White, black, blue, and red represent the TP pixels, TN pixels, FP pixels, and FN pixels, respectively. Gray represents the background pixels.

TABLE II
CD PERFORMANCE ON THE BAY AREA DATASET WITH 1% TRAINING SAMPLES

Method	OA(%)	AA(%)	$\kappa \times 100$	F1(%)	P(%)	R(%)
CVA	89.20±0.02	88.87±0.02	78.18±0.04	90.26±0.02	87.12±0.01	93.64±0.03
MAD	89.42±0.01	89.18± 0.01	78.66±0.02	90.34±0.01	88.19±0.02	92.60±0.01
SVM	93.92±0.16	94.10±0.20	87.83±0.33	94.14±0.14	97.05±0.91	91.41±0.79
GCN	96.61±0.40	96.59±0.31	93.19±0.80	96.83±0.43	96.74±0.89	96.93±1.67
D^2AGCN	97.04±0.10	97.05±0.10	94.06±0.20	97.22±0.09	97.52±0.16	96.93±0.04
CSDBF	98.13±0.37	98.18±0.36	96.25±0.74	98.24±0.35	**99.02±0.39**	97.47±0.70
D-LIEG	97.07±0.37	97.12±0.39	94.12±0.75	97.24±0.35	98.04±0.63	96.45±0.47
$SAGN_{base}$	97.74±0.77	97.76±0.78	95.47±1.55	97.88±0.72	98.18±0.75	97.59±0.71
SAGN	**98.46±0.61**	**98.47±0.62**	**96.90±1.23**	**98.55±0.57**	98.85±0.62	**98.26±0.54**

CSDBF achieves better performance compared to D^2AGCN and D-LIEG. This improvement can be attributed to CSDBF's consideration of the interaction between bitemporal HSIs, enabling the capture of interactive features from the two phases of HSIs. In the Bay Area scene, the SAGN again showcases highest CD indexes compared to other GNN-based methods, which proves the effectiveness of the LSGC once more. Compared to the $SAGN_{base}$, the accuracies of the SAGN have improved, indicating the significance of the dynamic graph learning strategy and joint loss in enhancing performance.

Fig. 4 shows the visual observations of all comparison methods on the Bay Area dataset. From the detection maps of CVA and MAD, it is evident that a substantial number of unchanged pixels have been misclassified into the changed class. The GNN-based methods, such as GCN, D^2AGCN, CSDBF, and D-LIEG, exhibit improved capability in distinguishing the unchanged pixels, but still produce some FN pixels. Consistent with the visual observations on the above dataset, the SAGN yields fewer misclassified pixels (including both FN pixels and FP pixels) compared to the $SAGN_{base}$, and shows improved CD visualization. Additionally, the SAGN demonstrates the capability to detect scattered changed regions that other GNN-based methods fail to identify. In general, SAGN exhibits better details and generates a satisfactory visual map.

3) Farmland Dataset: Table III reports the CD results of all compared methods on the Farmland dataset. The best CD accuracies are emphasized in boldface. CVA and MAD exhibit higher OA than the SVM and the GCN, which is inconsistent with the results observed on the other datasets. CVA and MAD can accurately classify the majority of changed pixels, but tend to misclassify a large number of unchanged pixels into changed class. D^2AGCN, and D-LIEG achieve better accuracies the GCN, yet they remain somewhat unsatisfactory. One potential explanation could be that the low-pass graph convolution might not be suitable for HSI CD. CSDBF performs

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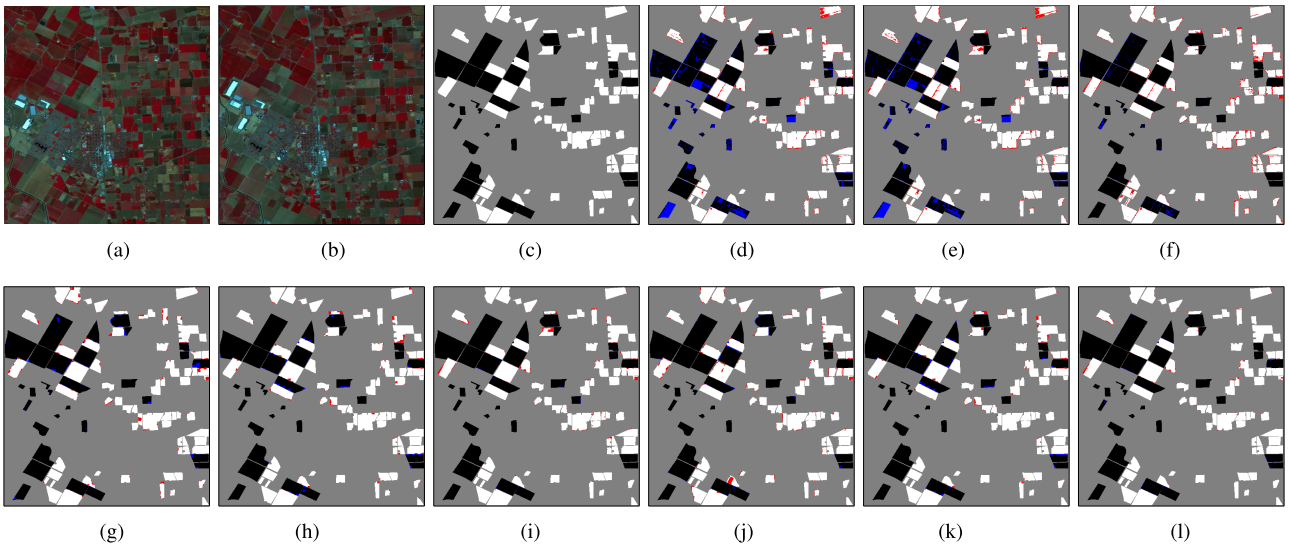


Fig. 4. CD maps of the compared methods on the Bay Area dataset. (a) False color image of Bay Area obtained in 2013. (b) False color image of Bay Area obtained in 2015. (c) Ground truth. (d) CVA, OA = 89.20%. (e) MAD, OA = 89.42%. (f) SVM, OA = 93.92%. (g) GCN, OA = 96.61%. (h) D^2AGCN, OA = 97.04%. (i) CSDBF, OA = 98.13%. (j) D-LIEG, OA = 97.07%. (k) $SAGN_{base}$, OA = 97.74%. (l) SAGN, OA = 98.46%. White, black, blue, and red represent the TP pixels, TN pixels, FP pixels, and FN pixels, respectively. Gray represents the background pixels.

TABLE III
CD PERFORMANCE ON THE FARMLAND DATASET WITH 1% TRAINING SAMPLES

Method	OA(%)	AA(%)	$\kappa \times 100$	F1(%)	P(%)	R(%)
CVA	94.95±0.01	94.64±0.00	88.03±0.01	91.63±0.01	89.89±0.02	93.92±0.01
MAD	94.81±0.01	94.73±0.01	87.72±0.02	91.43±0.01	88.89±0.02	94.54±0.01
SVM	93.94±0.09	92.62±0.16	85.24±0.17	89.50±0.10	89.70±0.73	89.46±0.67
GCN	93.56±0.54	92.89±0.40	84.59±1.21	89.17±0.82	87.16±1.49	91.28±0.12
D^2AGCN	94.44±0.17	94.08±0.58	86.73±0.47	90.68±0.37	88.27±1.03	93.23±1.71
CSDBF	95.99±0.50	94.89±0.31	90.21±1.15	93.02±0.79	**93.81±1.79**	92.28±0.46
D-LIEG	93.99±0.21	94.06±0.40	85.79±0.40	90.10±0.24	86.32±1.52	94.21±1.62
$SAGN_{base}$	95.73±0.33	94.97±0.32	89.67±0.79	92.68±0.55	92.20±0.82	93.17±0.27
SAGN	**96.75±0.38**	**96.50±0.29**	**92.18±0.87**	**94.48±0.59**	93.12±1.51	**95.89±0.78**

effectively in this dataset, because it proposes a complement strategy to aggregate both temporal and spatial information from bitemporal HSIs. The proposed SAGN, utilizing the LSGC, demonstrates superior performance compared to other GNN-based methods. Overall, our SAGN outperforms other methods across most performance indexes, including OA, AA, κ, F1, and R under 1% of training ratio.

Fig. 5 showcases the visual CD maps on the Farmland dataset. The CVA and MAD methods generate many misclassified unchanged pixels, i.e., FP pixels (blue). SVM has fewer misclassified unchanged pixels than CVA, but obviously contains many FN pixels (red) in the lower right part of the change map. The GCN, D^2AGCN, and CSDBF demonstrate a strong capability to differentiate the unchanged pixels, but tend to generate many FN pixels. In contrast, D-LIEG can correctly classify the majority of changed pixels, but misclassify a lot of unchanged pixels (bule). From the visual maps, our proposed SAGN presents the fewest FP pixels and FN pixels, resulting in the most visually optimal performance among the compared methods.

4) Computational Complexity: In this section, experiments are conducted to compare the computational complexity of different methods. As shown in Table IV, during the training stage, the SAGN spends more time than GCN, D^2AGCN, and $SAGN_{base}$. The main reason is that, in order to obtain more suitable graphs, the SAGN spends extra time to dynamically adjust graph structure in each layer, while graphs are fixed in the GCN, D^2AGCN, and $SAGN_{base}$. However, it is important to highlighted that the SAGN can greatly improve accuracy by obtaining dynamic graphs, but does not significantly increase the training time and the number of parameters. In contrast, although the CSDBF and D-LIEG can also achieve performance improvement, they usually contain larger numbers of parameters and spend much more training time than our

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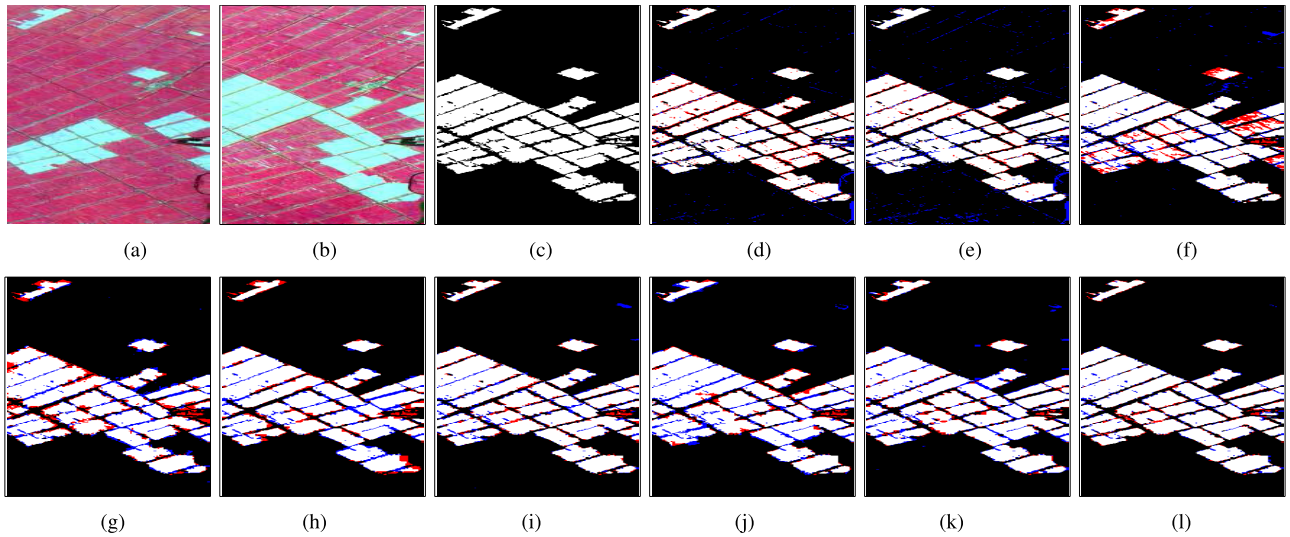


Fig. 5. CD maps of the compared methods on the Farmland dataset. (a) False color image of Farmland obtained in 2006. (b) False color image of Farmland obtained in 2007. (c) Ground truth. (d) CVA, OA = 94.95%. (e) MAD, OA = 94.81%. (f) SVM, OA = 93.94%. (g) GCN, OA = 93.56%. (h) D^2AGCN, OA = 94.44%. (i) CSDBF, OA = 95.99%. (j) D-LIEG, OA = 93.99%. (k) $SAGN_{base}$, OA = 95.73%. (l) SAGN, OA = 96.75%. White, black, blue, and red represent the TP pixels, TN pixels, FP pixels, and FN pixels, respectively.

TABLE IV
TRAINING TIME, TESTING TIME, AND NUMBER OF PARAMETERS ON THREE DATASETS

Methods	Santa Barbara			Bay Area			Farmland		
	Training time(s)	Testing time(s)	Parameters	Training time(s)	Testing time(s)	Parameters	Training time(s)	Testing time(s)	Parameters
GCN	1198.77	3.64	124930	1263.76	3.43	124930	115.48	0.14	107266
D^2AGCN	1245.12	3.75	125826	1791.18	5.47	125826	124.04	0.14	107886
CSDBF	5209.99	5.84	166082	3581.63	8.95	166082	145.61	0.31	139172
D-LIEG	4988.32	4.31	4827138	3637.88	5.70	4827138	156.45	0.12	4791810
$SAGN_{base}$	1256.12	3.78	124930	1561.33	4.42	124930	119.36	0.22	107266
SAGN	1603.36	5.02	157954	1812.87	5.60	157954	160.35	0.40	140290

TABLE V
ABLATION STUDY ON THE FARMLAND DATASET

Method	Module			Farmland					
	LSGC	Dynamic graph	Joint loss	OA(%)	AA(%)	$\kappa \times 100$	F1(%)	P(%)	R(%)
GCN	×	×	×	93.56±0.54	92.89±0.40	84.59±1.21	89.17±0.82	87.16±1.49	91.28±0.12
$SAGN_{base}$	✓	×	×	95.73±0.33	94.97±0.32	89.67±0.79	92.68±0.55	92.20±0.82	93.17±0.27
$SAGN_{DG}$	✓	✓	×	96.36±0.42	95.99±0.36	91.23±0.98	93.80±0.67	92.58±1.47	95.12±0.73
SAGN	✓	✓	✓	**96.75±0.38**	**96.50±0.29**	**92.18±0.87**	**94.48±0.59**	**93.12±1.51**	**95.89±0.78**

SAGN. In general, compared with other methods, the proposed SAGN is able to significantly improve the performance at the expense of some efficiency.

C. Ablation Study

In this section, we assess the impacts of the primary components of the proposed SAGN, namely the LSGC module, the dynamic graph strategy, and the joint loss.

1) Boosting Change Information Between Bitemporal HSIs With the LSGC Module: To verify the advantage and superiority of the proposed LSGC module, we compare the $SAGN_{base}$ with its corresponding Laplacian-smoothing-based variant, namely the GCN. The differences between the $SAGN_{base}$ and the GCN is that the GCN employs the Laplacian smoothing-based graph convolution at each layer, and relies on the "similarity graph" to guide the graph convolution. Table V reports the CD performance, with the highest accuracies highlighted in boldface. As depicted in Table V, $SAGN_{base}$ attains notably superior performance compared to the GCN, which demonstrates the significant advantage of the LSGC. This implies that, compared to the smoothing operation, enhancing the difference information proves to be more beneficial for HSI CD.

2) Improving Adaptability With the Dynamic Graph Strategy: We are intrigued by how the dynamic graph strategy contributes to HSI CD performance. For this, we construct a variant named $SAGN_{DG}$, which integrates both the LSGC Module and the dynamic graph strategy. Both models utilize identical hyperparameters throughout the experiments. The performance comparison is shown in Table V. Compared to

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$SAGN_{base}$, $SAGN_{DG}$ enhances the detection performance by learning refined graph structure. This suggests that dynamically capturing the graph structures of bitemporal HSIs is beneficial and improves performance.

3) Enhancing Discriminative Ability With the Joint Loss: To assess the efficacy of the joint loss, the variant model $SAGN_{CE}$ is constructed, which retains the same framework but is trained supervised by the cross-entropy loss. Table V reports the performance of the SAGN and the $SAGN_{CE}$. The SAGN consistently outperforms the $SAGN_{CE}$ across all the performance indexes. More importantly, the SAGN demonstrates notable accuracy in R. This signifies that the SAGN significantly enhances the detection performance of the changed class, which has fewer labeled samples available than unchanged class. This enhancement is attributed to the integration of the modified cross-entropy loss and contrastive loss, which enables the SAGN to prioritize and concentrate on the changed class, consequently improving its detection performance.

IV. Conclusion

This article proposes an SAGN tailored for HSI CD. First, leveraging the novel LSGC module, the SAGN effectively amplifies the difference information within changed regions, resulting in enhanced CD performance. Second, through constructing "difference graphs" and refining graph structures, the SAGN enhances its ability to extract crucial difference information from bitemporal HSIs. The distinguishable difference information facilitates the detection of both changed and unchanged pixels. Finally, the joint loss of the modified cross-entropy loss and contrastive loss improves the interclass discrimination and mitigates the challenges posed by imbalanced labeled samples in HSI CD. The comprehensive experimental results demonstrate that the SAGN attains state-of-the-art CD performance, exceling in both visualization quality and accuracy. Although the SAGN achieves superior performance for HSI CD, it can be further improved. In our future work, we propose to design a graph U-Net framework to model spatial topologies at multiple scales. Besides, we propose to construct an interaction graph and design an interactive framework to fully mine interaction information between bitemporal HSIs.

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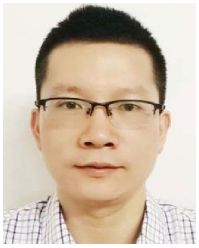
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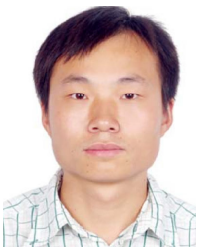
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